

Yu-Han Wu 吳雨翰

PhD student in diffusion models and machine learning theory
LPSM, Sorbonne University · Google DeepMind · Paris, France

EDUCATION

PhD in applied mathematics and machine learning	2024 – 2027 (expected)
Sorbonne University (LPSM) and Google DeepMind, Paris	
Advisors: Gérard Biau, Claire Boyer and Pierre Marion; Quentin Berthet and Romuald Elie at Google DeepMind.	
Diploma of the École Normale Supérieure, applied mathematics	2024
École Normale Supérieure, Paris	
MSc in statistics and machine learning	2023
Université Paris-Saclay	
BSc in applied mathematics	2021
Université Paris Cité	
Classe préparatoire (MPSI/MP)	2020
Lycée Louis-le-Grand, Paris	

PUBLICATIONS

CONFERENCE AND JOURNAL PAPERS

- Yu-Han Wu**, Quentin Berthet, Gérard Biau, Claire Boyer, Romuald Elie and Pierre Marion (2026). Optimal Stopping in Latent Diffusion Models. In International Conference on Machine Learning (ICML).
Oral at the PriGM workshop, EurIPS 2025
- Yu-Han Wu**, Pierre Marion, Gérard Biau and Claire Boyer (2025). Taking a Big Step: Large Learning Rates in Denoising Score Matching Prevent Memorization. In Proceedings of the Thirty Eighth Conference on Learning Theory (COLT), PMLR 291:5718–5756.
- Pierre Marion, **Yu-Han Wu**, Michael E. Sander and Gérard Biau (2024). Implicit regularization of deep residual networks towards neural ODEs. In The Twelfth International Conference on Learning Representations (ICLR).
Spotlight
- David Conlon and **Yu-Han Wu** (2023). More on lines in Euclidean Ramsey theory. Comptes Rendus. Mathématique, 361(G5), 897–901.

PREPRINTS

- Guillaume Couairon, Alexis Jacq, **Yu-Han Wu**, Renu Singh, Yana Hasson, Quentin Berthet and Romuald Elie (2026). Kastor: An efficient fine-tuning strategy for generative emulation of PDE simulations. arXiv:2608.06107.
- DiffusionGemma Team (2026). DiffusionGemma Technical Report. arXiv:2608.00146.
- Pierre Marion and **Yu-Han Wu** (2026). Understanding diffusion models requires rethinking (again) generalization. arXiv:2605.06077.
- Quentin Berthet, **Yu-Han Wu**, Clément Crepy, Romuald Elie, Klaus Greff and Michael E. Sander (2026). MIND: Monge Inception Distance for Generative Models Evaluation. arXiv:2605.06797.

TALKS AND POSTERS

Invited talk at Optimisation et Apprentissage appliqués aux contenus numériques	Jun 2026
MathSTIC, USPN, France	

Poster at Lemanth 2026 EPFL Bernoulli Center, Lausanne, Switzerland	Apr 2026
Oral at PriGM Workshop, EurIPS 2025 Copenhagen Bella Center, Copenhagen, Denmark	Dec 2025
Talk at COLT 2025 ENS Lyon, Lyon, France	Jul 2025
Talk at Journées de Statistique 2025 Université Aix-Marseille, Marseille, France	Jul 2025

DISTINCTIONS

Oral at the PriGM workshop, EurIPS 2025 Optimal Stopping in Latent Diffusion Model	2025
Spotlight, ICLR 2024 Implicit regularization of deep residual networks towards neural ODEs	2023

RESEARCH EXPERIENCE

Research intern Owkin, Paris Federated learning and invariant risk minimization. Supervisor Ulysse Marteau-Ferey.	Winter 2023
Research intern Sorbonne University (LPSM) Implicit regularization of residual networks. Supervisors Gérard Biau and Pierre Marion.	Summer 2023
Research intern California Institute of Technology Extremal combinatorics and Euclidean Ramsey theory. Supervisor David Conlon.	Summer 2022

SERVICE

Reviewer, ICML	2026
Reviewer, NeurIPS	2026
Reviewer, PriGM workshop at EurIPS	2025 – 2026

SKILLS

Mathematics	probability, statistics, optimization, stochastic analysis
Programming	Python (JAX, PyTorch, NumPy), LaTeX, Git